

Programmatic AI: A Paradigm for Interpretable, Transferable Machine Intelligence

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Argos Research is pursuing artificial intelligence (AI) systems that are controllable and come with behavioral guarantees (interpretable and verifiable), apply their experience to new scenarios (transfer-learning), exhibit compositional and conceptual understanding (hierarchical learning), and generalize robustly beyond their training environments (sim-to-real).

Current AI is dominated by end-to-end deep learning architectures (DL), enjoying tremendous success in various domains [1–4]. Despite this, DL suffers from enormous data requirements [5], an opaque nature [6], and often displays a lack of compositional reasoning [7, 8]. When trained in simulation (sim), DL tends towards representing statistical correlations specific to the sim, instead of re-usable abstractions [9, 10]. While mitigation tactics exist, an uncommon class of remedy reaches for program synthesis (PS) [11–14]. PS suffers from astronomical search spaces, and a difficulty with high-dimensional inputs. Successful attacks on large search spaces in PS center around neural guided search [15], but remain confined to low-dimensional input domains such as state vectors [16]. Conversely, hand-crafted domain specific languages (DSLs) let PS take high-dimensional inputs directly [17], but are underexplored in their synthesis methods and rely solely on genetic programming, struggling with local optima.

We propose ‘Programmatic AI’: a novel paradigm based on evolutionary program search in which reinforcement learning in simulation guides programmatic policy iteration, where the improvement operator combines a neural-guided search algorithm (NGSA) with an evolutionary algorithm (EA) over policy runtime fitness, applied to partial programs under a learned prior, in an initially hand-crafted, self-evolving DSL. This rests on novel NGSAs synthesizing programs that read high-dimensional inputs directly. The programmatic nature of the policy means policies are verifiable, while Cartesian Genetic Programming [18] based EAs allow us to side-step bloat [19], and avoid calculation of unconnected ‘genes’, greatly enhancing search efficiency, thereby making PS feasible. Empirically, programmatic policies tend towards generalizable algorithms [20]; which is reinforceable by scoring candidate programs across multiple tasks and sims. Bounded program length pressures a program against pathologically enumerating strategies specific to each sim or task. We expect sim-to-real and transfer-learning to follow from this regime as a whole: a programmatic policy, scored across tasks and simulations, under a length bound.

Challenges lie in the inherent complexity of desired behaviors across tasks. Programs assessed across tasks and sims risk reward exploitation for ill-defined reward functions. Poorly chosen DSLs, mutation structures, or search algorithms will induce complex, unoptimized code, straining compute and time budgets, exacerbated in our regime by executing programs directly on costly sims. However, if successful, physical AI, and indeed signal processing more broadly, can shed its reliance on huge, varied, expensive real-world data sets [21–24], its lack of hierarchical understanding, and its statistical brute-forcing, and enter an age of machine intelligence that adapts rather than memorizes. Deployments in novel and multi-agent conditions such as in defense, mining, agriculture, and forestry can now be trained for cheap in simulation, producing sophisticated, self-directed behavior.

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